Carbon & nanoparticles

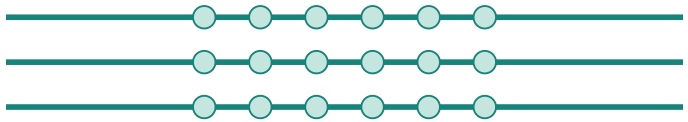
Link carbon bonds

1 Diamond

Each carbon bonds to four others in a rigid giant covalent network. Many strong bonds → very hard, high melting.

No delocalised electrons or moving ions → does not conduct electricity. Useful for cutting.

2 Graphite & graphene



Layers can slide past each other

Graphite: each carbon bonds to three others in hexagonal layers. One electron per carbon is delocalised → conducts.

Weak forces between layers → slide → lubricant. Strong bonds within layers → high melting.

Graphene: a single layer; very strong and conducting, useful in composites/electronics.

Apply structures

3 Fullerenes & nanotubes

Fullerenes are hollow carbon cages/tubes, based on hexagonal rings; some also have five- or seven-sided rings. C_{60} is a cage.

Large surface area: catalysts. Hollow cages: drug delivery. Cylindrical nanotubes: high tensile strength and conductivity → composites/electronics.

4 Size classes

Nanoparticles: **1–100 nm.** Fine particles: 100–2,500 nm. Coarse particles: 2,500–10,000 nm.

1 nm = 10^{-9} m. Smaller size increases surface area compared with volume; properties may differ from bulk material.

Assess tiny particles

5 Calculate the ratio

Cube of side x: area = $6x^2$; volume = x^3; **SA/V = 6/x.** Use the same length units.

Side 10 nm → SA/V 0.6 nm^{-1}; side 5 nm → 1.2 nm^{-1}. Halving size doubles the ratio.

More exposed surface per amount → less catalyst/coating may be needed.

6 Benefits & risks

Uses: catalysts, medicine, electronics, sunscreen and antibacterial coatings. Link each use to a useful property.

Small particles may enter the body; effects depend on material, dose and exposure route.

Compare benefit with toxicity/environmental evidence and uncertainties; do not assume all nanoparticles are safe or harmful.